

EFFECT OF PARTIAL STERILIZATION BY STEAM OR FORMALIN ON DAMPING-OFF OF SITKA SPRUCE

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(With Plate 11 and 1 Text-figure)

Three species of *Pythium* isolated from alkaline soil of an old established nursery at Ampthill were found to be parasitic to Sitka spruce seedlings. Symptoms varied from typical damping-off to a slow root-rot. Not all infected seedlings died, some surviving as stunted plants with partly diseased root systems. Pathogenic *Pythium* species were common in nursery soil at Ampthill, but absent from forest soil there, while pathogenic and non-pathogenic species were more abundant in the old than in the newer sections of the Kennington nursery. *P. ultimum* was abundant in soil from three alkaline nurseries (pH 6.8-7.2) and rare or absent in soil from two acid nurseries (pH 5.3-5.5).

The effect of partial sterilization of nursery soil at Ampthill was, in part, to destroy pathogens. *Pythium* did not survive either the initial steam or formalin treatment, and subsequent recolonization of treated soil was slow. Formalin was highly toxic to *Pythium* species in agar cultures and in soil. *Trichoderma viride*, which is more tolerant of formalin than most other soil fungi, was often the dominant recolonizer of formalin-treated soil.

Steam or formalin treatment markedly improved numbers and height of Sitka spruce seedlings in the first year after treatment and showed a moderate residual effect in the second year, but gave little improvement in the third year.

INTRODUCTION

During recent years Sitka spruce and other conifer seedlings have shown very poor growth in many old established forest nurseries in Great Britain. Fertilizers alone have had little effect on productivity, whereas partial sterilization has increased it (Crowther, 1950). Partial sterilization is known to affect the physical properties of soil, to increase the solubility of nutrients, and to cause profound changes in the soil microflora and fauna, but there is little information about the way in which these changes affect the growth of conifer seedlings.

The effect of partial sterilization by steam or formalin on the fungus flora of an old forest nursery soil was studied at the Forestry Commission's Old Ampthill Nursery, Bedfordshire, where the seedlings have been raised for about 25 years, Sitka spruce showing very poor growth in the more recent years. Seedlings from steamed or formalin-treated plots are taller, sturdier and have much larger root systems than plants from untreated plots. Warcup (1951*b*) showed that in this nursery soil fungi were initially destroyed or reduced to very low numbers by steam or formalin treatments and that subsequent fungal recolonization of treated soil was slow.

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The common agents of conifer damping-off hitherto described are *Pythium* species, *Corticium solani* (*Rhizoctonia solani*), with species of *Fusarium* and *Phytophthora* active in certain localities (Rathbun-Gravatt, 1931; Roth & Riker, 1943). *Fusarium* and *Pythium* species are common in Ampthill nursery soil (Warcup, 1951*b*), but little is known about the identity of the pathogens in Great Britain. Hartley (1921) and others have pointed out that damping-off in forest nursery beds is in many cases primarily a root-rot.

GROWTH OF SITKA SPRUCE SEEDLINGS IN THE NURSERY

During 1950, Sitka spruce (*Picea sitchensis* Carr.) seedlings grown in untreated plots at Ampthill (Warcup, 1951*b*) were examined for the occurrence of damping-off. Seed of Sitka spruce was sown on 20–21 March and germination commenced about a month later. Examination of seed-beds 3–7 weeks after germination showed a few seedlings with typical damping-off symptoms. A greater number, however, died gradually and stood upright, shrivelled and brown, in the nursery beds. The number of seedlings which had died increased until mid-July, by which time an estimated 70–80 % of the seedlings had died. Few seedlings died after July but those which survived remained small and were typical of Sitka spruce grown in a 'conifer sick' old established nursery. In May–June seedlings from sterilized soil had clean, white roots, whereas most of those from untreated soil showed all degrees of damage from small lesions to a chestnut-brown discoloration of the entire root system. Lesions were not confined to ground-level but occurred at various points throughout the length of the root. Discoloured roots were often shrivelled and decayed for a greater or lesser part of their length. Seedlings with extensively damaged roots usually showed a shrivelling of the stem before death. Older seedlings had secondary roots arising above the decayed root portions, often with subsequent damage to these secondary roots.

ISOLATION OF FUNGI FROM SITKA SPRUCE SEEDLINGS

Throughout the 1950 growing season fungi were isolated from living Sitka spruce seedlings grown in untreated plots. The whole roots were washed, sterilized for 3 min. in 1 in 1000 mercuric chloride, washed three times in sterile water and plated on Dox + yeast agar. Table 1 records the fungi most commonly isolated; the value of 100 for percentage occurrence means that a particular fungus was isolated from every root.

Table 1. *Percentage occurrence of fungal isolates from roots of Sitka spruce seedlings grown in untreated nursery plots*

Date of sampling	... 24 May	30 May	6 June	26 June	24 July	15 August
No. of seedlings examined...	22	82	80	100	55	25
Common fungi isolated:						
<i>Pythium</i> spp.	36	40	18	2	0	4
<i>Fusarium</i> spp.	5	27	53	100	55	72
<i>Cylindrocarpum radicola</i>	9	20	7	30	52	60
<i>Penicillium</i> spp.	9	0	5	3	37	16
<i>Chaetomium</i> spp.	9	2	6	3	22	28

Table 1 shows that *Pythium* species were commonly recorded in spring but were rarely isolated later in the year. Most of the other fungi isolated increased throughout the growing season. Three species of *Pythium* (*P. ultimum* Trow, *Pythium* E39 and *Pythium* E16) and two strains of *Fusarium oxysporum* sensu Snyder & Hansen were the fungi most commonly isolated, but other species of these genera were occasionally recorded.

PATHOGENICITY OF ISOLATES OF *PYTHIUM* AND *FUSARIUM*
TO SITKA SPRUCE SEEDLINGS

The pathogenicity to Sitka spruce of *Cylindrocarpon radicola* Wollenw., and of all species of *Pythium* and *Fusarium* isolated from seedlings or soil at Ampthill was tested in the laboratory. One-pound jars were filled with sterilized Kettering loam potting soil of pH 6.0. Seventeen cultures were tested. Three jars were allowed for each culture and six jars for controls. For inoculation, 0.6 cm. disks cut from the edges of actively growing fungal colonies upon Dox + yeast agar were used. One disk

Table 2. *Pathogenicity of isolates of Pythium, Fusarium and Cylindrocarpon to Sitka spruce seedlings*

Fungus	Emergence (%)	Post- emergence loss (%)	Total loss* (%)
<i>Pythium ultimum</i> Trow	65	94	95
<i>Pythium</i> E 16	63	85	88
<i>Pythium</i> E 16 strain b	68	56	62
<i>Pythium</i> E39	69	90	91
<i>Pythium</i> E39 strain b	67	97	98
<i>Pythium</i> E24	79	0	1
<i>Pythium</i> ? <i>acanthicum</i> Drechsler	85	0	0
<i>P. dissotocum</i> Drechsler	81	0	0
<i>P. rostratum</i> Butler	84	0	0
<i>P. intermedium</i> de Bary	77	0	4
<i>Pythium</i> E 150	87	0	0
<i>Fusarium oxysporum</i> strain 1	73	1	10
<i>F. oxysporum</i> strain 2	83	0	0
<i>F. culmorum</i> (W. G. Smith) Sacc.	82	0	0
<i>F. equiseti</i> (Corda) Sacc.	76	0	5
<i>F. sambucinum</i> Fuckel	80	0	0
<i>Cylindrocarpon radicola</i> Wollenw.	81	0	0
Control	80	0	0

* Expressed as percentage of seedlings emerging in controls.

was placed about 1 cm. below the level of the seed-bed in each jar. Each jar was sown with fifty seeds. The seeds were covered with a $\frac{3}{16}$ in. layer of a coarse, unsterilized sand, as the use of a coarse seed-bed cover is normal nursery practice. This sand had been tested and found to be free from *Pythium* and *Fusarium*. Throughout the experiment, the soil was maintained at 45 % of its water holding capacity by frequent weighing and by adding water as necessary. Damped-off seedlings were removed with sterile forceps at intervals up to a month. All seedlings plated on agar yielded the inoculated fungus. The results are given in Table 2. The experiment was later repeated with Ampthill soil when the *Fusarium* inoculum consisted of the fungus grown for 1 month on maize-meal-sand (3:100) medium. Two grams of inoculum were incorporated with the

soil at seed level in each jar. The results of this experiment were similar to those reported in Table 2, except that the percentage seed germination in soil inoculated with pathogenic species of *Pythium* was lower.

Three species of *Pythium* were thus highly pathogenic to Sitka spruce seedlings grown in steam-sterilized soil. The *Fusarium* strains and *Cylindrocarpum radiculicola* were apparently of little importance, for experiments with *Pythium* and *Fusarium* isolates inoculated together into soil showed that while both could be isolated from seedlings after 2-3 weeks no seedling yielded *Fusarium* alone, and the presence of *Fusarium* did not increase the severity of the disease.

From these and similar experiments it has been found that the *Pythium* species studied caused a wide range of disease symptoms in Sitka spruce, depending on the moisture content of the soil. *P. ultimum* was generally the most virulent species; at low soil moisture it caused a root-rot which gradually killed seedlings and left them standing upright. At high soil moisture infected seedlings showed typical damping-off symptoms and rotted rapidly. In sterilized Ampthill soil kept at 70 % of its water-holding capacity, *P. ultimum* caused pre-emergence losses ranging between

Table 3. *Damping-off of Sitka spruce seedlings grown in untreated Ampthill soil kept at different moisture contents*

	Moisture content as percentage water-holding capacity			
	15-20	35-40	55-60	75-80
Percentage emergence of seedlings	0*	64	76	26
Percentage post-emergence loss	0	84	93	100

* Seed healthy, soil too dry for germination.

70 and 100 %, all surviving seedlings being killed within 10-14 days of emergence. The symptoms caused by *Pythium* E16 also varied with soil moisture; in many of the infected seedlings, only the curved hypocotyls appeared above the ground. These curved seedlings often stayed green for appreciable periods, the stem being apparently resistant, though the root was attacked soon after emergence from the seed coat. *Pythium* E39 typically caused heavier post-emergence than pre-emergence losses irrespective of soil moisture, and in general attacked the seedlings at a later stage than the other two fungi (Pl. 11, figs. 3, 4), causing a root-rot.

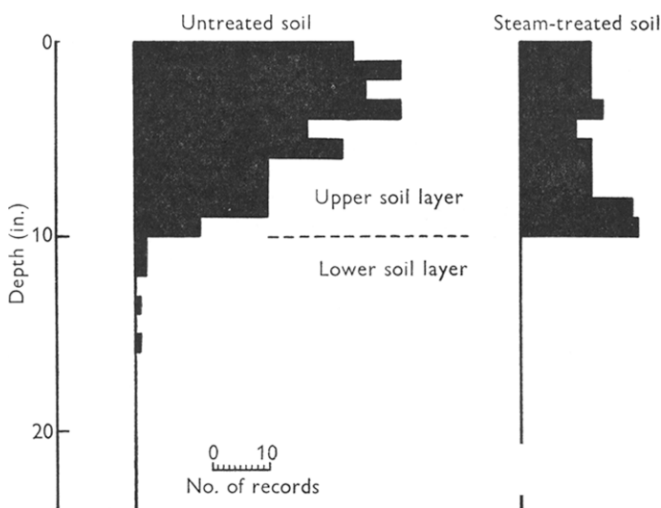
In pathogenicity experiments spruce seedlings which survived 1 month after inoculation were kept for a further 6 weeks. Deaths continued up to the end of that period. Surviving seedlings from *Pythium*-infected soil were significantly smaller than seedlings of the same age from uninfected controls. In seedlings from *Pythium*-infected soil the primary roots were generally decayed and secondary roots originated above the region of decay. A few secondary roots were also decayed. Height of seedlings was approximately inversely proportional to the length of root infected.

The loss caused by *Pythium* species present in untreated naturally contaminated Ampthill nursery soil is shown in Table 3, which records losses of seedlings in soil kept at different moisture levels in the laboratory. The seed used gave an 80 % germination in sterilized soil.

Even at comparatively low soil moisture contents damping-off losses, particularly post-emergence losses, were very heavy in naturally contaminated Ampthill nursery soil. This experiment has been repeated over the lower soil moisture range (35–40 % of water-holding capacity) with post-emergence losses ranging between 42 and 95 %.

OCCURRENCE OF *PYTHIUM* IN NURSERY SOIL

Warcup (1951*b*) showed that *Pythium* species (pathogenic and non-pathogenic) were widespread throughout the upper layer and rare in the lower layer of Ampthill nursery soil. An analysis of their occurrence on soil plates (Warcup, 1950) from soil samples collected over a period of 3 years showed that they were recorded most abundantly from a depth



Text-fig. 1. Occurrence of *Pythium* in profiles of untreated and recolonized steam-treated nursery soil.

of 2–4 in., and that below 6 in. their numbers fell off to a depth of 10–12 in., the limit of the upper soil layer (Text-fig. 1). Only four colonies of *Pythium* on some 540 soil plates were recorded from the lower soil layer, and the deepest level at which a *Pythium* colony was found was 16 in. There was some evidence that *Pythium* species occurred less frequently in the top inch or two of soil in summer than in winter. This accords with the findings of McLaughlin (1947). The species isolated most commonly was *Pythium* E16. *P. ultimum*, *Pythium* E39, *Pythium* E24, *P. intermedium* and *P. ? acanthicum* were isolated frequently, and other species were rare. *Pythium* E150 was recorded only from larger humus particles which had been washed free from other soil particles.

To find out whether *Pythium* species were present as spores or mycelium in nursery soil, a study of the source of *Pythium* colonies developing on soil plates was made by the method outlined by Warcup (1951*a*). Soil plates were prepared and incubated at 22.5° C. for 12–24 hr. and then examined for the presence of *Pythium* colonies. Widely spaced colonies coming from

arger soil particles were cultured by removing a hyphal tip to an agar slope. The remainder of the colony with a surrounding block of agar was cut out and placed on a microscope slide. Most of the agar around the colony was carefully cut away, the remaining agar and colony were stained with lactophenol-cotton blue diluted 1:10 with water, and a cover-slip added. Some care was necessary to avoid breaking the colony into pieces, but if the agar disintegrated under the weight of the cover-slip a good preparation usually resulted. Preparations were sealed with gum dammar-beeswax sealing compound (Dade & Waller, 1949).

Some 55 *Pythium* colonies were mounted on slides; others were examined without being made into preparations. Most colonies were isolated from soil from Tair Onen nursery, Glamorgan, Wales, but a few were from Ampthill. All the *Pythium* colonies had originated from spores, but whether from oospores, or resting sporangia was often difficult to decide. Oospores were sometimes found still surrounded by the oogonium wall (Pl. 11, fig. 1). Most spores were found to be surrounded by, and sometimes deeply embedded in, a mass of small humus and clay particles, the whole mass being irregular in outline (Pl. 11, figs. 1, 2). Unless a stained germ tube and spore contents could be seen, it was often difficult to determine whether such masses contained a *Pythium* spore or not.

Although it is reported that oospores of *Pythium* are difficult to germinate (Middleton, 1943), it is notable that in these experiments oospores germinated easily. *Pythium* spores are often surrounded by soil particles to form relatively large masses which may be the reason why *Pythium* species are not usually recorded in soil dilution studies (Chesters, 1949). High dilutions would also tend to obscure the presence of *Pythium* in soil.

Since *Pythium* species were of common occurrence in nursery soil, an examination was made of soil from the conifer forest at Ampthill. Soil samples were collected from the upper 6 in. of twelve profiles from different parts of the forest and the fungal floras of these profiles examined. *Pythium* species were found in a grass ride between forest sections, and in a section of pines grown after deciduous trees, a few of which were still present. They were not recorded from pure pine stands or under Douglas fir where the soil was more acid than elsewhere. All *Pythium* isolates recorded from forest sections were found to be non-pathogenic to Sitka spruce seedlings.

EFFECT OF PARTIAL STERILIZATION ON THE OCCURRENCE OF *PYTHIUM* SPECIES IN NURSERY SOIL

Both with steam and with formalin treatment, fungi in Ampthill nursery soil were initially destroyed or reduced to very low numbers (Warcup, 1951*b*). The penetration of soil was 16 in. for Hoddesdon steaming and 5 in. for formalin treatment. Subsequent fungal recolonization of treated soil was very slow and different under the different treatments. Steam-treated soil was recolonized chiefly by species of *Mortierella*, *Phoma eupyrena* and *Coniothyrium* in the first year, whereas *Trichoderma viride* was the dominant recolonizer of formalin-treated soil in the same period.

There was no evidence that *Pythium* species survived the initial partial sterilization treatments, and their recolonization of treated soil was

relatively slow. Table 4 records the occurrence of *Pythium* species at Ampthill over a period of 3 years after soil treatments with steam and formalin in March 1949. These records represent total *Pythium* (pathogenic and non-pathogenic) since most colonies were recorded without isolation. Occurrence throughout the year has been divided into two periods 'spring' (March to July) and 'autumn' (August to November). No samples were collected during December to February. Damping-off of Sitka spruce in the nursery occurred predominantly in the spring period of the year.

Pythium species were first recorded from steam-treated soil 13 weeks after treatment, and recolonization was very slight during the first spring. Their numbers rose in the autumn, remained at that level throughout the second spring, showed a further increase in the second autumn, and were at a level comparable with that recorded in untreated plots in the third spring after treatment. Not all of the colonies of *Pythium* from steamed plots were isolated and identified. *Pythium* E16, *P. ultimum* and *Pythium* E24, however, were isolated in the second year and six species, including

Table 4. *Effect of partial sterilization treatments on subsequent occurrence of Pythium in nursery soil*

Treatment—March 1949	Percentage of soil plates on which <i>Pythium</i> recorded				
	1949		1950		1951 Spring
	Spring	Autumn	Spring	Autumn	
Steam, Hoddesdon	0.5	8	10	33	41
Formalin, treated zone, 0–5 in.	0	0	3	6	—
Formalin, upper layer, 0–10 in.	15	11	21	17	20
Untreated	33	30	46	37	40

the three pathogenic species, were recorded in the third year after treatment. Damping-off was not noted in these steamed plots during 1949, was sporadic in 1950 and serious in 1951.

In a further series of nursery plots at Ampthill which were steamed by the Hoddesdon method in 1950, *Pythium* species were not recorded during the spring period and were present on only 2% of the plates during the autumn period.

By the end of the second year after treatment a difference was noted in the location of *Pythium* in profiles of steamed plots when contrasted with untreated plots. Whereas in untreated soil, *Pythium* species were most frequently recorded from 2 to 4 in. deep, in steamed soil they were most frequently isolated from 8 to 10 in. deep. Text-fig. 1 shows histograms of the numbers of plates from profiles of steamed and untreated soil from which *Pythium* species were recorded.

Table 4 shows that *Pythium* species were not recorded from treated soil until 12 months after the application of formalin. They were, however, still present in the untreated soil below the level of formalin penetration. During the second year their numbers remained low in the treated soil zone. At the end of the second year, the plots were cultivated to a depth of 7–8 in. so that the treated soil layer was broken up and the surface soil layers contained a mixture of untreated and formalin-treated soil.

In nursery plots treated with formalin in 1950, *Pythium* species were not recorded during the spring period and were present on 4% of the soil plates during the following autumn.

Table 5 records the growth of Sitka spruce seedlings on these plots over the period of 3 years after the soil treatments (assessments through the courtesy of the Forestry Commission). Partial sterilization treatment by steam or formalin greatly improved the height and number of seedlings in the first year, gave a considerable residual effect in the second year, and showed little improvement in the growth of Sitka spruce seedlings in the third year after treatment. Losses of seedlings throughout all plots were much greater in 1950 and 1951 than in 1949.

Table 5. *Effect of partial sterilization by steam or formalin on growth of Sitka spruce seedlings in old forest nursery soil*

	1949		1950		1951	
	Total no. of seed- lings per sq.yd.	Mean height in in.	Total no. of seed- lings per sq.yd.	Mean height in in.	Total no. of seed- lings per sq.yd.	Mean height in in.
Treatment, March 1949						
Steam (Hoddesdon)	801	2.46	241	1.6	182	0.48
Formalin	774	1.84	217	1.18	190	0.56
Untreated	373	1.03	60	0.97	86	0.34

EFFECT OF FORMALIN ON SOIL FUNGI

Formalin treatment of soil in the nursery at Ampthill increased *Trichoderma viride* (Warcup, 1951*b*) and decreased *Pythium*. The effect of different concentrations of formalin on soil fungi was, therefore, tested in soil and in agar cultures. Different concentrations of formalin were added to a Kettering loam potting soil and to the Ampthill soil in glass tumblers (Garrett, 1936) in the laboratory. Formalin applied at the rate of 0.1 gal. of commercial formalin per sq.yd. (made up in 1 gal. of water) killed all fungi in the top layers of Ampthill soil in field plots, so the equivalent amount (0.35 ml./sq.in.) was used as a standard, 'S', for dilution studies. Progressively reduced fractions of the standard in a specified volume of water (enough to bring air dry soil to 50% water-holding capacity) were added to 150 g. lots of air dry soil in glass tumblers. After addition of formalin, tumblers were closed with moisture-proof cellophane covers and left for 2 days. Tumblers were sampled at a depth of 1 in. by the soil-plate method, three plates being prepared from each tumbler. After sampling, tumblers were sown with cress seed to measure the toxicity of different concentrations of formalin and to note any damping-off of cress seedlings by *P. ultimum*. Table 6 records the results of these experiments which have been repeated with the same general results. The results were, however, rather variable and sometimes the same strength of formalin killed more fungi than on other occasions.

Table 6 shows that at certain concentrations formalin may act as a true 'partial' sterilizing agent, killing some fungi but not others. In these soils *Penicillium* species (*P. luteum*, *P. wortmanni* and *P. purpurogenum*) were usually the most resistant fungi to formalin. They were followed by

Trichoderma viride, *Fusarium* species, *Phoma eupyrena*, and *Botrytis* among others. *Pythium ultimum* appears very susceptible to formalin, since damping-off was prevented at formalin concentrations which left most other soil fungi still viable. In confirmation of a previous report by Haenseler (1935), Ogilvie, Hickman and Croxall (1938) found formalin diluted 1/200–1/600 would control damping-off of various seedlings if added after the seed was sown. Some seedling injury also occurred, especially to seedlings of *Matthiola* at the lower dilutions.

Table 6 shows that Kettering loam treated with certain concentrations of formalin was rapidly recolonized by *Trichoderma viride*, which spored profusely and visibly throughout the soil after 7–10 days at laboratory temperatures. Soil plates showed that *T. viride* also recolonized Amphill soil after the application of certain concentrations of formalin, but visible

Table 6. *Effect of different dilutions of formalin on occurrence of fungi in soil and on growth of cress seedlings*

Formalin dilution	Av. no. of species of fungi	Kettering loam			Amphill soil		
		Growth of cress	Visible fructifications of <i>Trichoderma viride</i>	Av. no. of species of fungi	Growth of cress	Visible fructifications of <i>Trichoderma viride</i>	
S	0	No germination	—	0	No germination	—	
S/4	0	No germination	—	0	No germination	—	
S/10	1	No germination	+++	0	No germination	—	
S/20	3	Seedling injury	+++	3	Seedling injury	—	
S/40	8	Seedling injury	++	6	Seedling injury	—	
S/60	11	Normal growth	+	10	Normal growth	—	
S/100	19	Normal growth	—	21	Normal growth	—	
S/200	22	Normal growth	—	29	Damping off	—	
Control	24	Normal growth	—	28	Damping off	—	

fructifications on soil were not observed. *T. viride* did not recolonize soil after the more concentrated or the more dilute doses of formalin. This may have been due either to the fact that formalin in high concentrations was toxic to the fungus or that *T. viride* had been killed throughout the tumblers. With dilute doses of formalin the abundance of *T. viride* decreased as the dilution of the solution increased and appeared inversely proportional to the number of other fungi present in the soil. It would appear that unless a certain proportion of the soil microflora was killed by the formalin *T. viride* did not abundantly recolonize the soil. Davey and Leach (1941), using formalin to control *Sclerotium rolfsii* in soil, found *Trichoderma* appearing upon considerable numbers of the sclerotia after formalin treatment. The percentage of sclerotia from which *Trichoderma* grew decreased as the dilution of the solution used for treating the soil increased. They supposed that *Trichoderma* invaded sclerotia killed or weakened by chemical treatment. Bliss (1951), in a study of the control of *Armillaria mellea* root-rot of citrus by partial soil sterilization with carbon disulphide, has correlated the death of *A. mellea* with the development of *Trichoderma viride* in soil after partial sterilization. Fumigation with CS₂ in sterilized soil did not kill *Armillaria mellea* in citrus root segments. *Trichoderma viride*, however, when grown in sterilized soil killed the fungus

even without fumigation. In roots in unsterilized soil containing *T. viride* and other fungi, *Armillaria mellea* remained viable when not fumigated, but was killed when fumigated, *Trichoderma viride* showing rapid development in the fumigated soil.

The effect of formalin on growth of soil fungi in agar culture was also examined. 1 ml. amounts of dilutions of formalin were placed in sterile Petri dishes, 10 ml. of cooled Dox + yeast agar (pH 5.0) were added and the plates then inoculated with 2 mm. squares of actively growing cultures of various soil fungi and incubated at 22.5° C. for 2-8 days. Table 7 shows the growth of each fungus recorded on an index of 0-10 where 10 is growth made on formalin-free agar.

Table 7. Growth in culture of soil fungi in presence of dilutions of formalin

Fungus	Dilutions of commercial formalin							
	1:100	1:250	1:500	1:750	1:1000	1:1500	1:2000	1:2500
<i>Pythium</i> E16	0	0	0	0	0	0	0	7
<i>P. intermedium</i>	0	0	0	0	0	0	0	2
<i>Mortierella</i> E23	0	0	0	0	0	0	0	5
<i>Mortierella</i> E31	0	0	0	0	0	0	1	—
<i>Pythium</i> E39	0	0	0	0	0	0	2	4
<i>Pythium</i> E24	0	0	0	0	0	0	4	4
<i>P. ultimum</i>	0	0	0	0	0	0	3	—
<i>Phytophthora erythroseptica</i>	0	0	0	0	0	0	8	7
<i>Chaetomium globosum</i>	0	0	0	1	3	5	6	—
<i>Phoma eupyrena</i>	0	0	9	10	10	10	10	10
<i>Cylindrocarpon radicola</i>	0	0	9	10	10	10	10	10
<i>Penicillium jenseni</i>	0	0	9	10	10	10	10	10
<i>P. nigricans</i>	0	2	7	10	10	10	10	—
<i>Botrytis cinerea</i>	0	1	10	10	10	10	10	—
<i>Fusarium equiseti</i>	0	6	9	10	10	10	10	—
<i>F. culmorum</i>	0	7	8	9	10	10	10	—
<i>F. oxysporum</i> strain 1	0	8	10	10	10	10	10	—
<i>Trichoderma viride</i>	0	0	3	10	10	10	10	—
<i>T. viride</i> B	1	8	9	10	10	10	10	—

Note: *Pythium dissotocum*, *Pythium* E150, *P. ? acanthicum*, and *P. rostratum* were inhibited by all dilutions of formalin. 1-50 formalin inhibited all fungi.

All *Pythium* species tested were sensitive to the presence of formalin in agar culture; other fungi such as *Fusarium* species and *Trichoderma viride* were much more tolerant. The results also show good agreement with the data obtained from nursery and laboratory experiments with fungi in soil. *Botrytis cinerea* showed an apparent stimulation of growth in agar containing 1:500 formalin.

Following upon these results, dilute formalin has been used to control *Pythium* which was masking the presence of other soil fungi developing from humus particles plated in agar. After dilute formalin treatment of the particles before plating, the number of plates showing *Pythium* colonies dropped from 10 in 12 to 1 in 12 with little apparent loss, and much easier recognition, of other soil fungi present in humus particles.

OCCURRENCE OF *PYTHIUM* SPECIES IN OTHER FOREST NURSERY SOILS

Since *Pythium* species were common in nursery soil at Amphill but rare in forest soil, soil samples from other forest nurseries were obtained in 1951 to investigate their *Pythium* content. Most samples were obtained from

old established nurseries or nursery sections where difficulty is experienced in growing Sitka spruce, though a few were from newer nursery sections in which growth was moderate or good. Soils were examined only for the occurrence of *Pythium* species.

Sections of different ages of a forest nursery soil from Kennington nursery, near Oxford, were examined. The oldest section (K 1/5) had been in use since 1925 and now gives poor growth of Sitka spruce, the second section (K 2D), from the nursery extension, had been in use since 1945 and shows moderate growth, and the third section, in which conifers had not yet been grown, K 3A, was agricultural soil recently incorporated into the nursery. Soil samples were collected some yards apart and from a depth of 2-4 in. on two different occasions during the summer, and counts made on the numbers of colonies of species of *Pythium* in 0.25 g. portions from each sample. Twenty soil plates from each sample were prepared using a total of 0.25 g. of soil, measured by weighing before and after preparation of soil plates, and the number of colonies of *Pythium* developing on these plates was counted. Plates were incubated at 22.5° C. and colonies counted after 1-4 days using a binocular dissecting microscope. Hyphal tips of colonies were tubed and the species determined. Table 8 records numbers of colonies of *Pythium* spp. isolated from the three sections. Samples 3-7 were collected on 22 June 1951, and samples 8 and 9 on 26 July 1951, the moisture contents of the latter samples being much lower than those of the former. Table 8 shows that the number of colonies and the number of species of *Pythium* were highest in the oldest section. This section was also alkaline whereas the newer sections were acid. *P. ultimum* was common in the old section, rare and sporadic in the nursery extension and was not isolated from the agricultural section. E 16, however, was much more abundant in the agricultural section than in the sections in which conifers have been grown. The pathogenicity of many of these isolates to Sitka spruce was determined and is shown in Table 10.

The *Pythium* species present in soil of other old established nurseries (Tair Onen, Wales; Tulliallen, Fife; Ringwood, Hants) which gave poor growth of Sitka spruce have been determined by the same method, and the results are given in Table 9. Numbers of colonies of different species were counted, but it is not known whether it is possible to correlate 'numbers of colonies' from different soils under different crops and with different moisture contents, in more than a general way. The number of counts made was small (3-4 except with Kennington), and it is possible that other species of *Pythium* are present in these soils but were not isolated. Since E 39 was isolated relatively more frequently from damped-off seedlings than from soil plates at Ampthill, it is possible that this or other pathogenic *Pythium* species might be isolated from diseased seedlings from these other nurseries.

Table 9 shows that Ampthill, the old section of Kennington (K 1/5), and Tair Onen nursery soils contain similar *Pythium* species, both pathogenic and non-pathogenic. These nurseries, or at least the parts investigated, are also all alkaline. Fewer species of *Pythium* were obtained from Old Ringwood and Tulliallen nurseries, which have acid soils, than from the alkaline soils in Ampthill, Kennington and Tair Onen. The lists from Old

Table 8. Numbers of colonies and species of *Pythium* isolated from soil from three sections of Kennington nursery

Nursery section ...	...	K1/5								K2D								K3A							
Age of section (years) ...	...	27								7								0							
Crop in 1950 and 1951 ...	...	Fallow								Fallow								Fallow							
pH of section (A.V.) ...	...	7.2								5.9								5.0							
Sample no. ...	...	3	4	5	6	7	8	9	Totals	3	4	5	6	7	8	9	Totals	3	4	5	6	7	8	9	Totals
Weight of soil used (g.) ...	...	0.26	0.25	0.24	0.27	0.25	0.25	0.26	1.78	0.25	0.25	0.28	0.26	0.25	0.27	0.25	1.80	0.25	0.26	0.25	0.26	0.27	0.26	0.25	1.80
<i>Pythium</i> E16*		13	14	9	5	4	7	3	55	5	4	5	3	5	9	3	34	15	17	21	21	25	7	7	113
<i>Pythium</i> E24		20	34	37	18	28	10	14	161	1	2	6	1	1	5	3	19	0	0	0	1	3	0	0	4
<i>P. intermedium</i>		4	13	5	6	1	3	2	34	4	7	12	2	4	1	1	31	2	0	0	0	0	1	0	3
<i>Pythium</i> K1. 1		19	11	5	11	3	1	5	55	1	1	4	0	0	0	0	6	0	0	0	0	0	0	0	0
<i>P. ? acanthicum</i>		0	0	5	4	10	6	11	36	0	0	0	0	0	2	3	5	0	0	0	0	0	0	2	2
<i>P. ultimum</i> *		15	9	6	8	18	19	12	87	0	2	0	0	1	1	0	4	0	0	0	0	0	0	0	0
<i>Pythium</i> K1.6		0	1	2	1	0	0	1	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
<i>Pythium</i> K1.14		1	0	0	1	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Totals		72	82	69	54	64	46	48	435	11	16	27	6	11	18	10	99	17	17	21	22	28	8		122

* Species pathogenic to Sitka spruce.

Table 9. Occurrence of *Pythium* species in forest nursery soils

Nursery section	...	...	Amphill	Kennington K1/5	Tair Onen	Tulli- allen	Ring- wood old	Kennington K2D
Crop in 1951	...	...	Sitka spruce	Fallow		Fallow	Fallow	Fallow
Av. pH of samples	...	...	7.1	7.2	7.2	5.3	5.5	5.9
% moisture content	...	...	12.0	15.8	19.5	18.8	9.4	15.0
<i>Pythium ultimum</i>			c.	c.	a.	—	r.	r.
<i>Pythium</i> E16			a.	c.	c.	a.	c.	c.
<i>Pythium</i> E39			c.	—	—	—	—	—
<i>Pythium</i> E24			c.	a.	c.	—	—	c.
<i>P. ? acanthicum</i>			c.	c.	r.	—	—	r.
<i>P. dissotocum</i>			r.	—	—	—	—	—
<i>P. rostratum</i>			r.	r.	r.	—	—	—
<i>P. intermedium</i>			c.	c.	c.	—	—	c.
<i>Pythium</i> E150			r.	—	—	—	—	—
<i>Pythium</i> K1.6			—	r.	r.	—	—	—
<i>Pythium</i> K1.1			—	c.	—	—	—	r.
<i>Pythium</i> Tul.1.11			—	—	—	r.	—	—
<i>Pythium</i> R1.11			—	—	—	—	c.	—
<i>P. mamillatum</i> Meurs			—	—	—	—	c.	—
<i>Pythium</i> R1.6			—	—	—	—	r.	—

a.=abundant; c.=common; r.=rare.

Table 10. Pathogenicity of *Pythium* species isolated from forest nursery soils to Sitka spruce seedlings

Experiment	...	...	A	B	C	
Soil	...	...	Kettering loam	Amphthill	Amphthill	
Soil moisture (% w.H.C.)	...	...	45	50	50	
			Sitka spruce	Sitka spruce	Sitka spruce	
			Post-emergence	Post-emergence	Post-emergence	
Fungus	Emergence	loss	Emergence	loss	Emergence	loss
	(%)	(%)	(%)	(%)	(%)	(%)
<i>Pythium ultimum</i> K 1	68	65	—	—	—	—
<i>P. ultimum</i> T	—	—	53	100	—	—
<i>P. ultimum</i> R	—	—	—	—	33	100
<i>Pythium</i> E 16 K 1	56	98	—	—	—	—
<i>Pythium</i> E 16 K 3	65	98	—	—	—	—
<i>Pythium</i> E 16 Tul.	—	—	57	95	—	—
<i>Pythium</i> E 16 R	—	—	—	—	50	78
<i>Pythium</i> E 16 T	—	—	31	90	—	—
<i>P. ? acanthicum</i> K 1	92	0	—	—	—	—
<i>P. ? acanthicum</i> T	—	—	83	0	—	—
<i>Pythium</i> E 24 K 1	85	0	—	—	—	—
<i>Pythium</i> E 24 T	—	—	86	0	—	—
<i>P. intermedium</i> K 1	87	0	—	—	—	—
<i>P. intermedium</i> T	—	—	85	2	—	—
<i>Pythium</i> K 1.6	91	0	—	—	—	—
<i>Pythium</i> K 1.1	93	1	—	—	—	—
<i>Pythium</i> K 1.14	93	0	—	—	—	—
<i>Pythium</i> T 5.1	—	—	—	—	62	3
<i>Pythium</i> R 1.6	—	—	—	—	68	0
<i>Pythium</i> Tul. 1.11	—	—	83	0	—	—
Control A	89	0	—	—	—	—
Control B	—	—	85	0	—	—
Control C	—	—	—	—	64	0

K1=Kennington K1/5; R=Ringwood; T=Tair Onen; K3=Kennington K3A; Tul.=Tulliallen.

Ringwood and Tair Onen also differed from one another. *P. ultimum* was rare or absent in the acid nursery soils, whereas it was common or abundant in the alkaline ones. Of these soils Tair Onen had the highest number of *Pythium* colonies, ranging from 595 to 687 per g., *P. ultimum* comprising from one-half to three-quarters of the total number. Counts for this soil were made using 0.10 g. of soil per twenty plates as it was difficult to count colonies with any accuracy in larger samples. The relative pathogenicities to Sitka spruce in sterilized soil of isolates of *Pythium* from these nurseries (except for two species from Ringwood) is given in Table 10.

Since *Pythium* E16 was found to be common in Kennington K3A, the agricultural soil which had not carried a conifer crop, the incidence of damping-off in untreated soil from the different sections of Kennington nursery was investigated. Untreated soil from Tair Onen and Tulliallen was also tested in the same experiment. Soils were adjusted to approximately 50 % of water-holding capacity and sown with Sitka spruce seed.

Table 11. *Percentage damping-off in untreated nursery soils*

Nursery section ...	... Kennington K1/5	Kennington K2D	Kennington K3A	Tair Onen	Tulliallen
Percentage emergence	77	90	90	38	90
Percentage post-emergence loss	74	6	3	86	6

The percentage germination and post-emergence losses over a period of 5 weeks are recorded in Table 11 which shows that damping-off was serious in Kennington K1/5 and Tair Onen, old established alkaline nursery soils containing *P. ultimum*. Damping-off was not serious in the other Kennington sections or at Tulliallen though E16, which is highly pathogenic in sterilized soil, was present in all three soils. It would appear that this species may be only mildly pathogenic under natural soil conditions or only become a serious parasite under certain environmental or nutritional conditions of the host.

This investigation was made possible by a grant kindly given by the Agricultural Research Council. Facilities for the field experiments were afforded through the courtesy of the Forestry Commission and they were carried out in co-operation with the Commission's Research Branch. I should like to express my thanks to Dr S. D. Garrett for continued encouragement, to Dr C. J. Hickman for the identification of *Pythium* isolates and to Mr F. T. N. Elborn for the preparation of photographs.

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EXPLANATION OF PLATE 11

- Fig. 1. Germinating oospore of *Pythium ultimum* within the oogonial wall, isolated from a soil plate. $\times 700$.
- Fig. 2. Germinating spore of *Pythium* sp. with attached soil particles isolated from a soil plate. $\times 700$.
- Fig. 3. Damping-off of Sitka spruce seedlings by different species of *Pythium* inoculated into steam-sterilized soil, 8 days after germination. E27 is *P. ultimum*.
- Fig. 4. Damping-off of Sitka spruce seedlings by different species of *Pythium* inoculated into steam-sterilized soil, 26 days after germination. E27 is *P. ultimum*.

(Accepted for publication 2 January 1952)

